

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Worked Solutions

Circles

Q001 Written

Find the equation of the circle with centre (3,-2) and radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 3)^2 + (y + 2)^2 = 25$.

Final answer

$$(x - 3)^2 + (y + 2)^2 = 25$$

Q002 Written

Find the equation of the circle with centre (-3,1) and radius 2.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x + 3)^2 + (y - 1)^2 = 4$.

Final answer

$$(x + 3)^2 + (y - 1)^2 = 4$$

Worked Solutions

Circles

Q003 Written

Find the equation of the circle with centre the origin and radius 4.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $x^2 + y^2 = 16$.

Final answer

$$x^2 + y^2 = 16$$

Q004 Written

Find the equation of the circle with centre (3,4), radius 6.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 3)^2 + (y - 4)^2 = 36$.

Final answer

$$(x - 3)^2 + (y - 4)^2 = 36$$

Worked Solutions

Circles

Q005 Written

Find the equation of the circle with centre $(-2, 6)$, radius $2\sqrt{3}$.**Solution**

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x + 2)^2 + (y - 6)^2 = 12$.

Final answer

$$(x + 2)^2 + (y - 6)^2 = 12$$

Q006 Written

Find the equation of the circle with centre the origin, radius 5.**Solution**

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $x^2 + y^2 = 25$.

Final answer

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 2)^2 = 25$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y + 2)^2 = 25$$

Q008 MCQ

Find the equation of the circle with centre (-3,1) and radius 2:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y - 1)^2 = 4$.

Correct option: A**Final answer**

$$(x + 3)^2 + (y - 1)^2 = 4$$

Worked Solutions

Circles

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 16$.

Correct option: D**Final answer**

$$x^2 + y^2 = 16$$

Q010 MCQ

Find the equation of the circle with centre (3,4), radius 6:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 4)^2 = 36$.

Correct option: B**Final answer**

$$(x - 3)^2 + (y - 4)^2 = 36$$

Worked Solutions

Circles

Q011 MCQ

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 6)^2 = 12$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 6)^2 = 12$$

Q012 MCQ

Find the equation of the circle with centre the origin, radius 5:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A**Final answer**

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q013 Written

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(3, -1)$, radius 2.

Final answer

circle; centre $(3, -1)$, radius 2

Q014 Written

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 3)$, radius 3.

Final answer

circle; centre $(2, 3)$, radius 3

Worked Solutions

Circles

Q015 Written

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, -4)$, radius 4.

Final answer

circle; centre $(2, -4)$, radius 4

Q016 Written

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 0)$, radius 2.

Final answer

circle; centre $(2, 0)$, radius 2

Worked Solutions

Circles

Q017 Written

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-1, 2)$, radius 5.

Final answer

circle; centre $(-1, 2)$, radius 5

Q018 Written

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Final answer

circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions

Circles

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(3, -1)$, radius 2.

Correct option: A**Final answer**circle; centre $(3, -1)$, radius 2

Worked Solutions

Circles

Q020 MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, 3)$, radius 3.

Correct option: C**Final answer**circle; centre $(2, 3)$, radius 3

Worked Solutions

Circles

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, -4)$, radius 4.

Correct option: D**Final answer**circle; centre $(2, -4)$, radius 4

Worked Solutions

Circles

Q022 MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, 0)$, radius 2.

Correct option: C**Final answer**circle; centre $(2, 0)$, radius 2

Worked Solutions

Circles

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(-1, 2)$, radius 5.

Correct option: C**Final answer**circle; centre $(-1, 2)$, radius 5

Worked Solutions

Circles

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Correct option: C**Final answer**circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions

Circles

Q025 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: centre (5, 3), radius² = 13; $(x - 5)^2 + (y - 3)^2 = 13$.

Final answer

centre (5, 3), radius² = 13; $(x - 5)^2 + (y - 3)^2 = 13$

Worked Solutions

Circles

Q026 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$.

Correct option: B

Final answer

centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

Worked Solutions

Circles

Q027 Written

Find the equation of the circle with centre (2,3) passing through (4,6).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y - 3)^2 = 13$.

Final answer

$$(x - 2)^2 + (y - 3)^2 = 13$$

Q028 Written

Find the equation of the circle with centre (-3,1) passing through (4,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 3)^2 + (y - 1)^2 = 58$.

Final answer

$$(x + 3)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q029 Written

Find the equation of the circle with centre (1,-3) through (-3,0).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 1)^2 + (y + 3)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 3)^2 = 25$$

Q030 Written

Find the equation of the circle with centre (-4,1) through (3,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 4)^2 + (y - 1)^2 = 58$.

Final answer

$$(x + 4)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q031 Written

Find the equation of the circle with centre (2,-1) through the origin.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y + 1)^2 = 5$.

Final answer

$$(x - 2)^2 + (y + 1)^2 = 5$$

Q032 Written

Find the equation of the circle with centre the origin through (4,-3).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 = 25$.

Final answer

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q033 MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y - 3)^2 = 13$.

Correct option: C**Final answer**

$$(x - 2)^2 + (y - 3)^2 = 13$$

Q034 MCQ

Find the equation of the circle with centre (-3,1) passing through (4,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y - 1)^2 = 58$.

Correct option: A**Final answer**

$$(x + 3)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y + 3)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y + 3)^2 = 25$$

Q036 MCQ

Find the equation of the circle with centre (-4,1) through (3,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 4)^2 + (y - 1)^2 = 58$.

Correct option: A**Final answer**

$$(x + 4)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 1)^2 = 5$.

Correct option: B**Final answer**

$$(x - 2)^2 + (y + 1)^2 = 5$$

Q038 MCQ

Find the equation of the circle with centre the origin through (4,-3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A**Final answer**

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q039 Written

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y + 1)^2 = 17$.

Final answer

$$(x - 2)^2 + (y + 1)^2 = 17$$

Q040 Written

Find the equation of the circle with diameter endpoints (3,4), (5,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 4)^2 + (y - 1)^2 = 10$.

Final answer

$$(x - 4)^2 + (y - 1)^2 = 10$$

Worked Solutions

Circles

Q041 Written

Find the equation of the circle with diameter endpoints (4,7), (-2,-1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 1)^2 + (y - 3)^2 = 25$.

Final answer

$$(x - 1)^2 + (y - 3)^2 = 25$$

Q042 Written

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 1)^2 + (y + 2)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 2)^2 = 25$$

Worked Solutions

Circles

Q043 Written

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 2)^2 + (y - 1)^2 = 2$.

Final answer

$$(x + 2)^2 + (y - 1)^2 = 2$$

Q044 Written

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + (y - 2)^2 = 25$.

Final answer

$$x^2 + (y - 2)^2 = 25$$

Worked Solutions

Circles

Q045 MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 1)^2 = 17$.

Correct option: A**Final answer**

$$(x - 2)^2 + (y + 1)^2 = 17$$

Q046 MCQ

Find the equation of the circle with diameter endpoints (3,4), (5,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y - 1)^2 = 10$.

Correct option: C**Final answer**

$$(x - 4)^2 + (y - 1)^2 = 10$$

Worked Solutions

Circles

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y - 3)^2 = 25$.

Correct option: C**Final answer**

$$(x - 1)^2 + (y - 3)^2 = 25$$

Q048 MCQ

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 25$.

Correct option: A**Final answer**

$$(x + 1)^2 + (y + 2)^2 = 25$$

Worked Solutions

Circles

Q049 MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 1)^2 = 2$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 1)^2 = 2$$

Q050 MCQ

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + (y - 2)^2 = 25$.

Correct option: C**Final answer**

$$x^2 + (y - 2)^2 = 25$$

Worked Solutions

Circles

Q051 Written

Find the equation of the circle passing through (1,1), (2,-1), and (3,2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 5x - y + 4 = 0$.

Final answer

$$x^2 + y^2 - 5x - y + 4 = 0$$

Q052 Written

Find the equation of the circle passing through (0,1), (-1,2), and (3,1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Final answer

$$x^2 + y^2 - 3x - 7y + 6 = 0$$

Worked Solutions

Circles

Q053 Written

Find the equation of the circle with through (0,0), (-1,1), (2,4).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x - 4y = 0$.

Final answer

$$x^2 + y^2 - 2x - 4y = 0$$

Q054 Written

Find the equation of the circle with through (-4,6), (0,8), (3,-1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 6y - 16 = 0$.

Final answer

$$x^2 + y^2 - 6y - 16 = 0$$

Worked Solutions

Circles

Q055 **Written****Find the equation of the circle with through (1,1), (5,-1), (-3,-7).****Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Final answer

$$x^2 + y^2 - 2x + 8y - 8 = 0$$

Q056 **MCQ****Find the equation of the circle passing through (1,1), (2,-1), and (3,2):****Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 5x - y + 4 = 0$.

Correct option: A**Final answer**

$$x^2 + y^2 - 5x - y + 4 = 0$$

Worked Solutions

Circles

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Correct option: B**Final answer**

$$x^2 + y^2 - 3x - 7y + 6 = 0$$

Q058 MCQ

Find the equation of the circle with through (0,0), (-1,1), (2,4):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x - 4y = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 2x - 4y = 0$$

Worked Solutions

Circles

Q059 MCQ

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6y - 16 = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 6y - 16 = 0$$

Q060 MCQ

Find the equation of the circle with through $(1,1)$, $(5,-1)$, $(-3,-7)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 2x + 8y - 8 = 0$$

Worked Solutions

Circles

Q061 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$.

Final answer

$$x^2 + y^2 - 6x - 6y + 8 = 0; \text{ centre } (3, 3), \text{ radius } \sqrt{10}$$

Worked Solutions

Circles

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$.

Correct option: A

Final answer

$$x^2 + y^2 - 6x - 6y + 8 = 0; \text{ centre } (3, 3), \text{ radius } \sqrt{10}$$

Worked Solutions

Circles

Q063 Written

Find the circle with centre (4,-2) tangent to the x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 4)^2 + (y + 2)^2 = 4$.

Final answer

$$(x - 4)^2 + (y + 2)^2 = 4$$

Q064 Written

Find the circle with centre (-2,3) tangent to the y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 2)^2 + (y - 3)^2 = 4$.

Final answer

$$(x + 2)^2 + (y - 3)^2 = 4$$

Worked Solutions

Circles

Q065 Written

Find the circle with centre $(-1,4)$ tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 4)^2 = 16$.

Final answer

$$(x + 1)^2 + (y - 4)^2 = 16$$

Q066 Written

Find the circle with centre $(3,-4)$ tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 3)^2 + (y + 4)^2 = 16$.

Final answer

$$(x - 3)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q067 Written

Find the circle with centre (5,-4) tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 5)^2 + (y + 4)^2 = 16$.

Final answer

$$(x - 5)^2 + (y + 4)^2 = 16$$

Q068 Written

Find the circle with centre (-1,6) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 6)^2 = 1$.

Final answer

$$(x + 1)^2 + (y - 6)^2 = 1$$

Worked Solutions

Circles

Q069 Written

Find the circle with centre (2,-5) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 2)^2 + (y + 5)^2 = 4$.

Final answer

$$(x - 2)^2 + (y + 5)^2 = 4$$

Q070 Written

Find the circle with centre (-3,-4) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 3)^2 + (y + 4)^2 = 9$.

Final answer

$$(x + 3)^2 + (y + 4)^2 = 9$$

Worked Solutions

Circles

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y + 2)^2 = 4$.

Correct option: B**Final answer**

$$(x - 4)^2 + (y + 2)^2 = 4$$

Worked Solutions

Circles

Q072 MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 3)^2 = 4$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 3)^2 = 4$$

Worked Solutions

Circles

Q073 MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 4)^2 = 16$.

Correct option: B**Final answer**

$$(x + 1)^2 + (y - 4)^2 = 16$$

Worked Solutions

Circles

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 4)^2 = 16$.

Correct option: D**Final answer**

$$(x - 3)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 5)^2 + (y + 4)^2 = 16$.

Correct option: D**Final answer**

$$(x - 5)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q076 MCQ

Find the circle with centre $(-1,6)$ tangent to y -axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 6)^2 = 1$.

Correct option: D**Final answer**

$$(x + 1)^2 + (y - 6)^2 = 1$$

Worked Solutions

Circles

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 5)^2 = 4$.

Correct option: B**Final answer**

$$(x - 2)^2 + (y + 5)^2 = 4$$

Worked Solutions

Circles

Q078 MCQ

Find the circle with centre $(-3,-4)$ tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y + 4)^2 = 9$.

Correct option: C**Final answer**

$$(x + 3)^2 + (y + 4)^2 = 9$$

Worked Solutions

Circles

Q079 Written

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y - 1)^2 = 1 \text{ or } (x + 5)^2 + (y - 5)^2 = 25$$

Q080 Written

Find the circles through $(1,2)$ tangent to both axes.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$.

Final answer

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ or } (x - 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q081 **Written****Find the circle with through $(-1, -2)$ tangent to both axes.****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Q082 **Written****Find the circle with through $(-2, -1)$ tangent to both axes.****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q083 Written

Find the circle with through (8,-1) tangent to both axes.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$.

Final answer

$$(x - 5)^2 + (y + 5)^2 = 25 \text{ or } (x - 13)^2 + (y + 13)^2 = 169$$

Worked Solutions

Circles

Q084 MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y - 1)^2 = 1 \text{ or } (x + 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q085 MCQ

Find the circles through (1,2) tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ or } (x - 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q086 MCQ

Find the circle with through $(-1,-2)$ tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q087 MCQ

Find the circle with through $(-2,-1)$ tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$.

Correct option: B**Final answer**

$$(x - 5)^2 + (y + 5)^2 = 25 \text{ or } (x - 13)^2 + (y + 13)^2 = 169$$

Worked Solutions

Circles

Q089 Written

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 1)^2 + (y + 1)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 1)^2 = 25$$

Q090 Written

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Final answer

$$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$$

Worked Solutions

Circles

Q091 **Written****Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 2)^2 + (y - 3)^2 = 2$.

Final answer

$$(x - 2)^2 + (y - 3)^2 = 2$$

Q092 **Written****Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3).****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 4)^2 + (y - 2)^2 = 26$.

Final answer

$$(x - 4)^2 + (y - 2)^2 = 26$$

Worked Solutions

Circles

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y + 1)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y + 1)^2 = 25$$

Worked Solutions

Circles

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Correct option: C**Final answer**

$$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$$

Worked Solutions

Circles

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y - 3)^2 = 2$.

Correct option: C**Final answer**

$$(x - 2)^2 + (y - 3)^2 = 2$$

Worked Solutions

Circles

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y - 2)^2 = 26$.

Correct option: D**Final answer**

$$(x - 4)^2 + (y - 2)^2 = 26$$

Worked Solutions

Circles

Q097 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Final answer

$$(x + 1)^2 + y^2 = 5; \text{ centre } (-1, 0), \text{ radius } \sqrt{5}$$

Worked Solutions

Circles

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Correct option: C

Final answer

$(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

Worked Solutions

Circles

Q099 Written

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.**Solution**

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1$ or $k > 3$.

Final answer

$$k < -1 \text{ or } k > 3$$

Q100 Written

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.**Solution**

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Final answer

$$k < -\sqrt{6} \text{ or } k > \sqrt{6}$$

Worked Solutions

Circles

Q101 Written

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < 5/4$.

Final answer

$$k < 5/4$$

Q102 Written

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1/5$ or $k > 3$.

Final answer

$$k < -1/5 \text{ or } k > 3$$

Worked Solutions

Circles

Q103 Written

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $-1 < k < 4$.

Final answer

$$-1 < k < 4$$

Worked Solutions

Circles

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1$ or $k > 3$.

Correct option: C

Final answer

$$k < -1 \text{ or } k > 3$$

Worked Solutions

Circles

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Correct option: B**Final answer**

$$k < -\sqrt{6} \text{ or } k > \sqrt{6}$$

Worked Solutions

Circles

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < 5/4$.

Correct option: C**Final answer**

$$k < 5/4$$

Worked Solutions

Circles

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1/5$ or $k > 3$.

Correct option: B**Final answer**

$$k < -1/5 \text{ or } k > 3$$

Worked Solutions

Circles

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-1 < k < 4$.

Correct option: D

Final answer

$$-1 < k < 4$$

Worked Solutions

Circles

Q109 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < k < 4$.

Final answer

$$0 < k < 4$$

Worked Solutions

Circles

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < k < 4$.

Correct option: B

Final answer

$$0 < k < 4$$

Worked Solutions

Circles

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(1, -1)$ and $(-1, 3)$.

Final answer

$(1, -1)$ and $(-1, 3)$

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(0, 2)$ and $(2, 0)$.

Final answer

$(0, 2)$ and $(2, 0)$

Worked Solutions

Circles

Q113 Written

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(3, -1)$.

Final answer $(3, -1)$

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$.

Final answer $(-2, 1)$

Worked Solutions

Circles

Q115 Written

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$ and $(4, -1)$.

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions

Circles

Q116 MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(1, -1)$ and $(-1, 3)$.

Correct option: C**Final answer** $(1, -1)$ and $(-1, 3)$

Worked Solutions

Circles

Q117 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is (0, 2) and (2, 0).

Correct option: D**Final answer**

(0, 2) and (2, 0)

Worked Solutions

Circles

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(3, -1)$.

Correct option: C**Final answer** $(3, -1)$

Worked Solutions

Circles

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: D**Final answer** $(-2, 1)$

Worked Solutions

Circles

Q120 MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$ and $(4, -1)$.

Correct option: B

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions

Circles

Q121 Written

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 0 points.

Final answer

0 points

Q122 Written

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 2 points : (1, -3) and (3, 1).

Final answer

2 points : (1, -3) and (3, 1)

Worked Solutions

Circles

Q123 **Written****Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.****Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$.

Final answer2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$ Q124 **Written****Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.****Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 0 points.

Final answer

0 points

Worked Solutions

Circles

Q125 Written

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 1 point : $(-2, 0)$.

Final answer

1 point : $(-2, 0)$

Worked Solutions

Circles

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: D**Final answer**

0 points

Worked Solutions

Circles

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2 points : (1, -3) and (3, 1).

Correct option: A**Final answer**

2 points : (1, -3) and (3, 1)

Worked Solutions

Circles

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$.

Correct option: C**Final answer**2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$

Worked Solutions

Circles

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: C**Final answer**

0 points

Worked Solutions

Circles

Q130 MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 1 point : $(-2, 0)$.

Correct option: D**Final answer**1 point : $(-2, 0)$

Worked Solutions

Circles

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 < k < 5$.

Final answer

$$-5 < k < 5$$

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{5}$.

Final answer

$$r = \sqrt{5}$$

Worked Solutions

Circles

Q133 Written

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm\sqrt{3}$.

Final answer

$$m = \pm\sqrt{3}$$

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0 < r < \sqrt{10}$.

Final answer

$$0 < r < \sqrt{10}$$

Worked Solutions

Circles

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm 4$.

Final answer

$$m = \pm 4$$

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $n < -10$ or $n > 10$.

Final answer

$$n < -10 \text{ or } n > 10$$

Worked Solutions

Circles

Q137 Written

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 \leq m \leq 5$.

Final answer

$$-5 \leq m \leq 5$$

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{2}$.

Final answer

$$r = \sqrt{2}$$

Worked Solutions

Circles

Q139 Written

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r > \sqrt{17}$.

Final answer

$$r > \sqrt{17}$$

Worked Solutions

Circles

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 < k < 5$.

Correct option: D

Final answer

$$-5 < k < 5$$

Worked Solutions

Circles

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{5}$.

Correct option: B

Final answer

$$r = \sqrt{5}$$

Worked Solutions

Circles

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm\sqrt{3}$.

Correct option: B

Final answer

$$m = \pm\sqrt{3}$$

Worked Solutions

Circles

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < \sqrt{10}$.

Correct option: B

Final answer

$$0 < r < \sqrt{10}$$

Worked Solutions

Circles

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm 4$.

Correct option: C

Final answer

$$m = \pm 4$$

Worked Solutions

Circles

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $n < -10$ or $n > 10$.

Correct option: A

Final answer

$$n < -10 \text{ or } n > 10$$

Worked Solutions

Circles

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 \leq m \leq 5$.

Correct option: C

Final answer

$$-5 \leq m \leq 5$$

Worked Solutions

Circles

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{2}$.

Correct option: B**Final answer**

$$r = \sqrt{2}$$

Worked Solutions

Circles

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r > \sqrt{17}$.

Correct option: D

Final answer

$$r > \sqrt{17}$$

Worked Solutions

Circles

Q149 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r .

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < r < 2$.

Final answer

$$0 < r < 2$$

Worked Solutions

Circles

Q150 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < 2$.

Correct option: D

Final answer

$$0 < r < 2$$

Worked Solutions

Circles

Q151 Written

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $\sqrt{14}$.

Final answer $\sqrt{14}$

Q152 Written

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 4.

Final answer

4

Worked Solutions

Circles

Q153 Written

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $2\sqrt{5}$.

Final answer

$$2\sqrt{5}$$

Q154 Written

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $3\sqrt{2}$.

Final answer

$$3\sqrt{2}$$

Worked Solutions

Circles

Q155 **Written****Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.****Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 2.

Final answer

2

Q156 **MCQ****Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:****Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\sqrt{14}$.

Correct option: A**Final answer** $\sqrt{14}$

Worked Solutions

Circles

Q157 MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 4.

Correct option: C**Final answer**

4

Q158 MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: D**Final answer** $2\sqrt{5}$

Worked Solutions

Circles

Q159 MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3\sqrt{2}$.

Correct option: B**Final answer** $3\sqrt{2}$

Q160 MCQ

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2.

Correct option: D**Final answer**

2

Worked Solutions

Circles

Q161 Written

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$$k = \pm 2$$

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm\sqrt{31}$.

Final answer

$$k = \pm\sqrt{31}$$

Worked Solutions

Circles

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$$k = \pm 2$$

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = 0$ or $k = -2$.

Final answer

$$k = 0 \text{ or } k = -2$$

Worked Solutions

Circles

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: C

Final answer

$$k = \pm 2$$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm\sqrt{31}$.

Correct option: D

Final answer

$$k = \pm\sqrt{31}$$

Worked Solutions

Circles

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: A

Final answer

$$k = \pm 2$$

Worked Solutions

Circles

Q168 MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = 0$ or $k = -2$.

Correct option: B

Final answer

$$k = 0 \text{ or } k = -2$$

Worked Solutions

Circles

Q169 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $k = -1$ or $k = 7$.

Final answer

$$k = -1 \text{ or } k = 7$$

Worked Solutions

Circles

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = -1$ or $k = 7$.

Correct option: A

Final answer

$$k = -1 \text{ or } k = 7$$

Worked Solutions

Circles

Q171 Written

Find the tangent to $x^2 + y^2 = 25$ at (4,3).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $4x + 3y = 25$.

Final answer

$$4x + 3y = 25$$

Q172 Written

Find the tangent to $x^2 + y^2 = 34$ at (5,3).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $5x + 3y = 34$.

Final answer

$$5x + 3y = 34$$

Worked Solutions

Circles

Q173 Written

Find the tangents $x^2 + y^2 = 10$ at (3,-1).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x - y = 10$.

Final answer

$$3x - y = 10$$

Q174 Written

Find the tangents $x^2 + y^2 = 29$ at (-5,-2).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-5x - 2y = 29$.

Final answer

$$-5x - 2y = 29$$

Worked Solutions

Circles

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$.

Final answer

$$-2x + y = 5$$

Worked Solutions

Circles

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $4x + 3y = 25$.

Correct option: A**Final answer**

$$4x + 3y = 25$$

Worked Solutions

Circles

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $5x + 3y = 34$.

Correct option: C**Final answer**

$$5x + 3y = 34$$

Worked Solutions

Circles

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3x - y = 10$.

Correct option: A**Final answer**

$$3x - y = 10$$

Worked Solutions

Circles

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5x - 2y = 29$.

Correct option: C**Final answer**

$$-5x - 2y = 29$$

Worked Solutions

Circles

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$.

Correct option: A**Final answer**

$$-2x + y = 5$$

Worked Solutions

Circles

Q181 Written

Find the tangents from (1,3) to $x^2 + y^2 = 5$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $2x + y = 5$ and $-x + 2y = 5$.

Final answer

$$2x + y = 5 \text{ and } -x + 2y = 5$$

Q182 Written

Find the tangents from (-2,4) to $x^2 + y^2 = 10$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-3x + y = 10$ and $x + 3y = 10$.

Final answer

$$-3x + y = 10 \text{ and } x + 3y = 10$$

Worked Solutions

Circles

Q183 Written

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$ and $x + 2y = 5$.

Final answer

$$-2x + y = 5 \text{ and } x + 2y = 5$$

Q184 Written

Find the tangents from $(2,1)$ to $x^2 + y^2 = 1$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $y = 1$ and $4x - 3y = 5$.

Final answer

$$y = 1 \text{ and } 4x - 3y = 5$$

Worked Solutions

Circles

Q185 Written

Find the tangents from (2,6) to $x^2 + y^2 = 20$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-x + 2y = 10$ and $2x + y = 10$.

Final answer

$$-x + 2y = 10 \text{ and } 2x + y = 10$$

Worked Solutions

Circles

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x + y = 5$ and $-x + 2y = 5$.

Correct option: D**Final answer**

$$2x + y = 5 \text{ and } -x + 2y = 5$$

Worked Solutions

Circles

Q187 MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-3x + y = 10$ and $x + 3y = 10$.

Correct option: A**Final answer**

$$-3x + y = 10 \text{ and } x + 3y = 10$$

Worked Solutions

Circles

Q188 MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$ and $x + 2y = 5$.

Correct option: D**Final answer**

$$-2x + y = 5 \text{ and } x + 2y = 5$$

Worked Solutions

Circles

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $y = 1$ and $4x - 3y = 5$.

Correct option: B**Final answer**

$$y = 1 \text{ and } 4x - 3y = 5$$

Worked Solutions

Circles

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ and $2x + y = 10$.

Correct option: C**Final answer**

$$-x + 2y = 10 \text{ and } 2x + y = 10$$

Worked Solutions

Circles

Q191 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions

Circles

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Correct option: B

Final answer

$$-x + 2y = 10 \text{ at } (-2, 4); 2x + y = 10 \text{ at } (4, 2)$$

Worked Solutions

Circles

Q193 Written

Classify $x^2 + y^2 = 4$ **and** $(x - 4)^2 + y^2 = 9$.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: intersect at 2 points.

Final answer

intersect at 2 points

Q194 Written

Classify $x^2 + y^2 = 9$ **and** $(x - 3)^2 + (y - 4)^2 = 4$ **from A-E**.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: B : touch externally.

Final answer B : touch externally

Worked Solutions

Circles

Q195 **Written****Classify** $(x - 1)^2 + y^2 = 5^2$ **and** $x^2 + y^2 + 2x - 6y = 3$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: E : intersect at 2 points.

Final answer E : intersect at 2 pointsQ196 **Written****Classify** $x^2 + y^2 = 4$ **and** $(x - 1)^2 + y^2 = 1$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: C : touch internally.

Final answer C : touch internally

Worked Solutions

Circles

Q197 Written

Classify $x^2 + y^2 = 18$ **and** $(x + 2)^2 + (y - 2)^2 = 2$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: C : touch internally.

Final answer C : touch internally

Q198 Written

Classify $(x + 3)^2 + y^2 = 16$ **and** $x^2 + y^2 - 2x - 8y - 8 = 0$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: E : intersect at 2 points.

Final answer E : intersect at 2 points

Worked Solutions

Circles

Q199 **Written****Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: D : one is inside the other .

Final answer D : one is inside the other

Worked Solutions

Circles

Q200 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$, the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is intersect at 2 points.

Correct option: C

Final answer

intersect at 2 points

Worked Solutions

Circles

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is B : touch externally.

Correct option: D

Final answer

B : touch externally

Worked Solutions

Circles

Q202 MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is E : intersect at 2 points.

Correct option: B**Final answer** E : intersect at 2 points

Worked Solutions

Circles

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: A

Final answer

C : touch internally

Worked Solutions

Circles

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: D**Final answer****C : touch internally**

Worked Solutions

Circles

Q205 MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is E : intersect at 2 points.

Correct option: B

Final answer

E : intersect at 2 points

Worked Solutions

Circles

Q206 MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is D : one is inside the other .

Correct option: C

Final answer

D : one is inside the other

Worked Solutions

Circles

Q207 **Written****Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x - 3)^2 + (y - 1)^2 = 45$.

Final answer

$$(x - 3)^2 + (y - 1)^2 = 45$$

Q208 **Written****Find the circle with centre (-3,-9) touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x + 3)^2 + (y + 9)^2 = 16$.

Final answer

$$(x + 3)^2 + (y + 9)^2 = 16$$

Worked Solutions

Circles

Q209 Written

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x - 3)^2 + (y + 9)^2 = 144$.

Final answer

$$(x - 3)^2 + (y + 9)^2 = 144$$

Worked Solutions

Circles

Q210 MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 1)^2 = 45$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y - 1)^2 = 45$$

Worked Solutions

Circles

Q211 MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y + 9)^2 = 16$.

Correct option: D**Final answer**

$$(x + 3)^2 + (y + 9)^2 = 16$$

Worked Solutions

Circles

Q212 MCQ

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 9)^2 = 144$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y + 9)^2 = 144$$

Worked Solutions

Circles

Q213 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + (y + 2)^2 = 9$.

Final answer

$$(x + 1)^2 + (y + 2)^2 = 9$$

Worked Solutions

Circles

Q214 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 9$.

Correct option: D

Final answer

$$(x + 1)^2 + (y + 2)^2 = 9$$

Worked Solutions

Circles

Q215 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$).

Final answer

$$2x - 4y - 5 = 0 \text{ (equivalently } y = x/2 - 5/4)$$

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x + y - 1 = 0$.

Final answer

$$x + y - 1 = 0$$

Worked Solutions

Circles

Q217 **Written****Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.****Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $12x - 6y - 7 = 0$.

Final answer

$$12x - 6y - 7 = 0$$

Q218 **Written****Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.****Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x = 1/2$.

Final answer

$$x = 1/2$$

Worked Solutions

Circles

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$).

Correct option: A**Final answer**

$$2x - 4y - 5 = 0 \text{ (equivalently } y = x/2 - 5/4)$$

Worked Solutions

Circles

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: C**Final answer**

$$x + y - 1 = 0$$

Worked Solutions

Circles

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $12x - 6y - 7 = 0$.

Correct option: B**Final answer**

$$12x - 6y - 7 = 0$$

Worked Solutions

Circles

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x = 1/2$.

Correct option: D**Final answer**

$$x = 1/2$$

Worked Solutions

Circles

Q223 Written

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(-3, -1)$ and $(-1, 3)$.

Final answer $(-3, -1)$ and $(-1, 3)$

Q224 Written

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(0, 2)$ and $(8/5, -6/5)$.

Final answer $(0, 2)$ and $(8/5, -6/5)$

Worked Solutions

Circles

Q225 **Written****Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.****Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: (1, 2) and (2, 1).

Final answer

(1, 2) and (2, 1)

Worked Solutions

Circles

Q226 MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-3, -1)$ and $(-1, 3)$.

Correct option: B**Final answer** $(-3, -1)$ and $(-1, 3)$

Worked Solutions

Circles

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(0, 2)$ and $(8/5, -6/5)$.

Correct option: C**Final answer** $(0, 2)$ and $(8/5, -6/5)$

Worked Solutions

Circles

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is (1, 2) and (2, 1).

Correct option: B**Final answer**

(1, 2) and (2, 1)

Worked Solutions

Circles

Q229 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(-4, 3)$ and $(3, -4)$.

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions

Circles

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-4, 3)$ and $(3, -4)$.

Correct option: C

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions

Circles

Q231 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 2x + 6y - 15 = 0$.

Final answer

$$x^2 + y^2 + 2x + 6y - 15 = 0$$

Q232 Written

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 4x + 2y = 0$.

Final answer

$$x^2 + y^2 + 4x + 2y = 0$$

Worked Solutions

Circles

Q233 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 10x + 20y - 39 = 0$.

Final answer

$$x^2 + y^2 + 10x + 20y - 39 = 0$$

Q234 Written

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 24x + 32y - 5 = 0$.

Final answer

$$x^2 + y^2 + 24x + 32y - 5 = 0$$

Worked Solutions

Circles

Q235 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 - 10x - 6y + 24 = 0$.

Final answer

$$x^2 + y^2 - 10x - 6y + 24 = 0$$

Worked Solutions

Circles

Q236 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x + 6y - 15 = 0$.

Correct option: D

Final answer

$$x^2 + y^2 + 2x + 6y - 15 = 0$$

Worked Solutions

Circles

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 4x + 2y = 0$.

Correct option: A

Final answer

$$x^2 + y^2 + 4x + 2y = 0$$

Worked Solutions

Circles

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 10x + 20y - 39 = 0$.

Correct option: D

Final answer

$$x^2 + y^2 + 10x + 20y - 39 = 0$$

Worked Solutions

Circles

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 24x + 32y - 5 = 0$.

Correct option: C

Final answer

$$x^2 + y^2 + 24x + 32y - 5 = 0$$

Worked Solutions

Circles

Q240 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 10x - 6y + 24 = 0$.

Correct option: B

Final answer

$$x^2 + y^2 - 10x - 6y + 24 = 0$$

Worked Solutions

Circles

Q241 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Final answer

$$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$$

Worked Solutions

Circles

Q242 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Correct option: A

Final answer

$$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$$

Worked Solutions

Circles

Quiz MCQ 1 **Concept Quiz · MCQ**

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre $(2, -3)$, radius 5.

Correct option: B

Final answer

centre $(2, -3)$, radius 5

Worked Solutions

Circles

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0.

Correct option: D

Final answer

0

Worked Solutions

Circles

Quiz MCQ 3 Concept Quiz · MCQ

Determine the chord length for radius r and centre-to-chord distance d :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{r^2 - d^2}$.

Correct option: A

Final answer

$$2\sqrt{r^2 - d^2}$$

Worked Solutions

Circles

Quiz MCQ 4 **Concept Quiz · MCQ**

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is externally tangent.

Correct option: D

Final answer

externally tangent

Worked Solutions

Circles

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre (1,-2) passing through (4,2).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 1)^2 + (y + 2)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 2)^2 = 25$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(4, 3)$, $(-4, 3)$.

Final answer

$$(4, 3), (-4, 3)$$

Worked Solutions

Circles

Quiz WRITTEN 7 Concept Quiz · Written

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x + 4y = 25$.

Final answer

$$3x + 4y = 25$$

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $\lambda = -1/2$; $x^2 + y^2 + 2x + 2y - 6 = 0$.

Final answer

$$\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$$

Answer key

Use the question number shown on each page.

Q001 $(x - 3)^2 + (y + 2)^2 = 25$	Q027 $(x - 2)^2 + (y - 3)^2 = 13$	Q053 $x^2 + y^2 - 2x - 4y = 0$	Q079 $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$
Q002 $(x + 3)^2 + (y - 1)^2 = 4$	Q028 $(x + 3)^2 + (y - 1)^2 = 58$	Q054 $x^2 + y^2 - 6y - 16 = 0$	Q080 $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$
Q003 $x^2 + y^2 = 16$	Q029 $(x - 1)^2 + (y + 3)^2 = 25$	Q055 $x^2 + y^2 - 2x + 8y - 8 = 0$	Q081 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q004 $(x - 3)^2 + (y - 4)^2 = 36$	Q030 $(x + 4)^2 + (y - 1)^2 = 58$	Q056 A	Q082 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q005 $(x + 2)^2 + (y - 6)^2 = 12$	Q031 $(x - 2)^2 + (y + 1)^2 = 5$	Q057 B	Q083 $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$
Q006 $x^2 + y^2 = 25$	Q032 $x^2 + y^2 = 25$	Q058 D	Q084 C
Q007 A	Q033 C	Q059 D	Q085 D
Q008 A	Q034 A	Q060 D	Q086 C
Q009 D	Q035 D	Q061 $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$	Q087 C
Q010 B	Q036 A	Q062 A	Q088 B
Q011 B	Q037 B	Q063 $(x - 4)^2 + (y + 2)^2 = 4$	Q089 $(x - 1)^2 + (y + 1)^2 = 25$
Q012 A	Q038 A	Q064 $(x + 2)^2 + (y - 3)^2 = 4$	Q090 $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
Q013 circle; centre (3, -1), radius 2	Q039 $(x - 2)^2 + (y + 1)^2 = 17$	Q065 $(x + 1)^2 + (y - 4)^2 = 16$	Q091 $(x - 2)^2 + (y - 3)^2 = 2$
Q014 circle; centre (2, 3), radius 3	Q040 $(x - 4)^2 + (y - 1)^2 = 10$	Q066 $(x - 3)^2 + (y + 4)^2 = 16$	Q092 $(x - 4)^2 + (y - 2)^2 = 26$
Q015 circle; centre (2, -4), radius 4	Q041 $(x - 1)^2 + (y - 3)^2 = 25$	Q067 $(x - 5)^2 + (y + 4)^2 = 16$	Q093 D
Q016 circle; centre (2, 0), radius 2	Q042 $(x + 1)^2 + (y + 2)^2 = 25$	Q068 $(x + 1)^2 + (y - 6)^2 = 1$	Q094 C
Q017 circle; centre (-1, 2), radius 5	Q043 $(x + 2)^2 + (y - 1)^2 = 2$	Q069 $(x - 2)^2 + (y + 5)^2 = 4$	Q095 C
Q018 circle; centre (-2, 3), radius $\sqrt{13}$	Q044 $x^2 + (y - 2)^2 = 25$	Q070 $(x + 3)^2 + (y + 4)^2 = 9$	Q096 D
Q019 A	Q045 A	Q071 B	Q097 $(x + 1)^2 + y^2 = 5$; centre (-1, 0), radius $\sqrt{5}$
Q020 C	Q046 C	Q072 B	Q098 C
Q021 D	Q047 C	Q073 B	Q099 $k < -1$ or $k > 3$
Q022 C	Q048 A	Q074 D	Q100 $k < -\sqrt{6}$ or $k > \sqrt{6}$
Q023 C	Q049 B	Q075 D	Q101 $k < 5/4$
Q024 C	Q050 C	Q076 D	Q102 $k < -1/5$ or $k > 3$
Q025 centre (5, 3), radius ² = 13; $(x - 5)^2 + (y - 3)^2 = 13$	Q051 $x^2 + y^2 - 5x - y + 4 = 0$	Q077 B	Q103 $-1 < k < 4$
Q026 B	Q052 $x^2 + y^2 - 3x - 7y + 6 = 0$	Q078 C	Q104 C

Q105 B	Q137 $-5 \leq m \leq 5$	Q169 $k = -1$ or $k = 7$	Q201 D
Q106 C	Q138 $r = \sqrt{2}$	Q170 A	Q202 B
Q107 B	Q139 $r > \sqrt{17}$	Q171 $4x + 3y = 25$	Q203 A
Q108 D	Q140 D	Q172 $5x + 3y = 34$	Q204 D
Q109 $0 < k < 4$	Q141 B	Q173 $3x - y = 10$	Q205 B
Q110 B	Q142 B	Q174 $-5x - 2y = 29$	Q206 C
Q111 (1, -1) and (-1, 3)	Q143 B	Q175 $-2x + y = 5$	Q207 $(x - 3)^2 + (y - 1)^2 = 45$
Q112 (0, 2) and (2, 0)	Q144 C	Q176 A	Q208 $(x + 3)^2 + (y + 9)^2 = 16$
Q113 (3, -1)	Q145 A	Q177 C	Q209 $(x - 3)^2 + (y + 9)^2 = 144$
Q114 (-2, 1)	Q146 C	Q178 A	Q210 A
Q115 (-2, 1) and (4, -1)	Q147 B	Q179 C	Q211 D
Q116 C	Q148 D	Q180 A	Q212 A
Q117 D	Q149 $0 < r < 2$	Q181 $2x + y = 5$ and $-x + 2y = 5$	Q213 $(x + 1)^2 + (y + 2)^2 = 9$
Q118 C	Q150 D	Q182 $-3x + y = 10$ and $x + 3y = 10$	Q214 D
Q119 D	Q151 $\sqrt{14}$	Q183 $-2x + y = 5$ and $x + 2y = 5$	Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)
Q120 B	Q152 4	Q184 $y = 1$ and $4x - 3y = 5$	Q216 $x + y - 1 = 0$
Q121 0 points	Q153 $2\sqrt{5}$	Q185 $-x + 2y = 10$ and $2x + y = 10$	Q217 $12x - 6y - 7 = 0$
Q122 2 points : (1, -3) and (3, 1)	Q154 $3\sqrt{2}$	Q186 D	Q218 $x = 1/2$
Q123 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$	Q155 2	Q187 A	Q219 A
Q124 0 points	Q156 A	Q188 D	Q220 C
Q125 1 point : (-2, 0)	Q157 C	Q189 B	Q221 B
Q126 D	Q158 D	Q190 C	Q222 D
Q127 A	Q159 B	Q191 $-x + 2y = 10$ at (-2, 4); $2x + y = 10$ at (4, 0)	Q223 (-3, -1) and (-1, 3)
Q128 C	Q160 D	Q192 B	Q224 (0, 2) and $(8/5, -6/5)$
Q129 C	Q161 $k = \pm 2$	Q193 intersect at 2 points	Q225 (1, 2) and (2, 1)
Q130 D	Q162 $k = \pm\sqrt{31}$	Q194 B : touch externally	Q226 B
Q131 $-5 < k < 5$	Q163 $k = \pm 2$	Q195 E : intersect at 2 points	Q227 C
Q132 $r = \sqrt{5}$	Q164 $k = 0$ or $k = -2$	Q196 C : touch internally	Q228 B
Q133 $m = \pm\sqrt{3}$	Q165 C	Q197 C : touch internally	Q229 (-4, 3) and (3, -4)
Q134 $0 < r < \sqrt{10}$	Q166 D	Q198 E : intersect at 2 points	Q230 C
Q135 $m = \pm 4$	Q167 A	Q199 D : one is inside the other	Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$
Q136 $n < -10$ or $n > 10$	Q168 B	Q200 C	Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q236 D**Q237** A**Q238** D**Q239** C**Q240** B

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Q242 A**Quiz MCQ 1** B**Quiz MCQ 2** D**Quiz MCQ 3** A**Quiz MCQ 4** D

Quiz WRITTEN 5 $(x - 1)^2 + (y + 2)^2 = 25$

Quiz WRITTEN 6 $(4, 3), (-4, 3)$

Quiz WRITTEN 7 $3x + 4y = 25$

Quiz WRITTEN 8 $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$